

Rimányi Chern-Positivity for A_5 : Verification, an ℓ -Free Reduction, and Three Dead Ends

Working report / handoff document

28 July 2026

Abstract

This report records an independent machine verification of the note `rimanyi_positivity.pdf` (Section 5 in particular), together with two results that are proved and three proof strategies that are ruled out. The headline: *Rimányi’s weak Chern-positivity conjecture for A_5 is not proved here.* What is proved is (i) a reduction showing that the conjecture “for all relative dimensions ℓ ” is a single ℓ -free statement indexed by zero-sum multisets, with the statements nested in ℓ ; and (ii) an exhaustive exact verification for $\ell = 0, \dots, 11$. This document is written to be self-contained enough to serve as the starting prompt for a subsequent working session.

Contents

1	Setup and conventions	1
1.1	The Bérczi–Szenes residue formula	1
1.2	Chamber expansion and the α/β dictionary	2
2	The two conjectures	2
3	Independent verification of the note	3
4	Proved: the ℓ-free reduction	3
5	Proved: exhaustive verification for $\ell \leq 11$	4
6	Closed form for F_5 and a maximal positive factorization	5
7	Three strategies that are ruled out	6
7.1	A $d = 4$ -style factorization proof cannot work	6
7.2	“Enumerate the negatives and dominate them” does not terminate	6
7.3	No closed form for $C(M)$ on simple families	6
8	The open problem, and what a proof would need	7
9	Caveats	7
10	Code	8

1 Setup and conventions

1.1 The Bérczi–Szenes residue formula

Let A_d be the Morin singularity of order d and let $\ell = k - n$ be the relative dimension of the map germ $(\mathbb{C}^n, 0) \rightarrow (\mathbb{C}^k, 0)$. The Thom polynomial is the iterated residue

$$\mathrm{Tp}_{A_d}^{n \rightarrow k} = \mathrm{Res}_{z_1=\infty} \cdots \mathrm{Res}_{z_d=\infty} \frac{(-1)^d \prod_{1 \leq m < l \leq d} (z_m - z_l) \mathcal{Q}_d(z_1, \dots, z_d)}{\prod_{l=1}^d \prod_{m=1}^{l-1} \prod_{r=1}^{\min(m, l-m)} (z_m + z_r - z_l)} \prod_{l=1}^d c\left(\frac{1}{z_l}\right) z_l^{k-n} dz_l, \quad (1)$$

where $c(t) = 1 + c_1 t + c_2 t^2 + \cdots$ is the total relative Chern class, $c_i = c_i(f^*TY - TX)$, with the conventions $c_0 = 1$ and $c_j = 0$ for $j < 0$.

The polynomial $\mathcal{Q}_d$ is the T_d -equivariant multidegree (equivalently the equivariant Poincaré dual) of the closure $\mathcal{O}_d = \overline{B_d} \cdot \widehat{e}$ of a Borel orbit inside the B_d -invariant subspace $\widehat{N}_d = \bigoplus_{1 \leq m+r \leq l \leq d} \mathbb{C} q_l^{mr} \subset \mathrm{Hom}(\mathbb{C}^d, \mathrm{Sym}^2 \mathbb{C}^d)$, where q_l^{mr} has T_d -weight $z_m + z_r - z_l$ and $\widehat{e} = \sum_{m=1}^d \sum_{r=1}^{d-m} q_{m+r}^{mr}$. It is homogeneous of degree $\mathrm{codim}_{\widehat{N}_d} \mathcal{O}_d$. The values used throughout, taken from the note, are

$$\mathcal{Q}_4 = 2z_1 + z_2 - z_4, \quad (2)$$

$$\mathcal{Q}_5 = (2z_1 + z_2 - z_5) P_5, \quad P_5 = 2z_1^2 + 3z_1 z_2 - 2z_1 z_5 + 2z_2 z_3 - z_2 z_4 - z_2 z_5 - z_3 z_4 + z_4 z_5. \quad (3)$$

Remark 1 (degree check). Homogeneity of the rational part of (1) forces $\deg \mathcal{Q}_d = \#\{\text{denominator factors}\} - \binom{d}{2}$, i.e. $\deg \mathcal{Q}_4 = 7 - 6 = 1$ and $\deg \mathcal{Q}_5 = 13 - 10 = 3$. Both (2) and (3) satisfy this. This is the only independent check on $\mathcal{Q}_d$ performed here; see Section 9.

1.2 Chamber expansion and the α/β dictionary

Under the convention for residues at infinity, expanding the denominators in the chamber $|z_1| \ll |z_2| \ll \cdots \ll |z_d|$ absorbs the sign $(-1)^d$, so the positivity question is read off from the chamber expansion of

$$R_d(z_1, \dots, z_d) = \frac{\prod_{1 \leq m < l \leq d} (z_m - z_l) \mathcal{Q}_d(z_1, \dots, z_d)}{\prod_{l=1}^d \prod_{m=1}^{l-1} \prod_{r=1}^{\min(m, l-m)} (z_m + z_r - z_l)}. \quad (4)$$

R_d is homogeneous of degree 0, so we may normalise $z_d = 1$ and set

$$x_j = \frac{z_j}{z_{j+1}} \quad (1 \leq j \leq d-1), \quad \text{equivalently} \quad z_l = x_l x_{l+1} \cdots x_{d-1}, \quad z_d = 1.$$

For $d = 5$ we write $(x_1, x_2, x_3, x_4) = (a, b, c, e)$, so that

$$z_1 = abce, \quad z_2 = bce, \quad z_3 = ce, \quad z_4 = e, \quad z_5 = 1, \quad (5)$$

and the chamber becomes $|a|, |b|, |c|, |e| \ll 1$. Write the chamber expansion as

$$F_d(x) = R_d = \sum_{\beta \in \mathbb{Z}_{\geq 0}^{d-1}} A_\beta x^\beta. \quad (6)$$

A monomial x^β equals z^α with

$$\alpha = (\alpha_1, \dots, \alpha_d) = (\beta_1, \beta_2 - \beta_1, \dots, \beta_{d-1} - \beta_{d-2}, -\beta_{d-1}), \quad (7)$$

and the extraction in (1) turns it into the contribution

$$A_\beta \prod_{i=1}^d c_{\ell+1+\alpha_i}, \quad \ell = k - n. \quad (8)$$

Lemma 2 (dictionary). $\beta \mapsto \alpha(\beta)$ is a bijection

$$\mathbb{Z}_{\geq 0}^{d-1} \xrightarrow{\sim} \left\{ \alpha \in \mathbb{Z}^d : \sum_i \alpha_i = 0, \quad \alpha_1 + \cdots + \alpha_j \geq 0 \text{ for } 1 \leq j \leq d-1 \right\}.$$

Proof. Immediate from (7): $\beta_j = \alpha_1 + \cdots + \alpha_j$ recovers β from α , the constraint $\beta_j \geq 0$ is exactly nonnegativity of the partial sums, and $\sum_i \alpha_i = 0$ is automatic. $\square$

Note $\sum_i (\ell + 1 + \alpha_i) = d(\ell + 1)$, the correct codimension of A_d .

2 The two conjectures

Conjecture 3 (Rimányi, weak Chern positivity). For every d and every ℓ , all coefficients of $\text{Tp}_{A_d}^{n \rightarrow k}$ expanded as a polynomial in the relative Chern classes c_i are nonnegative.

Conjecture 4 (stronger Laurent positivity). The chamber expansion $F_d = \sum_{\beta} A_{\beta} x^{\beta}$ of (4) has $A_{\beta} \geq 0$ for all β .

Conjecture 4 $\Rightarrow$ Conjecture 3 by (8), since each Chern coefficient is a sum of A_{β} 's. The converse fails, because several Laurent monomials feed the same Chern monomial. The note proves Conjecture 4 for $d = 4$ and disproves it for $d = 5$.

3 Independent verification of the note

All computations below were redone from scratch from (4), (2), (3) in exact integer arithmetic (`thom_positivity.py`; see Section 10).

Calibration. At $d = 4$, $\ell = 0$ the computation returns

$$\text{Tp}_{A_4}^{\ell=0} = c_1^4 + 6c_1^2c_2 + 2c_2^2 + 9c_1c_3 + 6c_4,$$

which is the classical value, with no global sign discrepancy. At $d = 5$, $\ell = 0$ it returns

$$\text{Tp}_{A_5}^{\ell=0} = c_1^5 + 10c_1^3c_2 + 10c_1c_2^2 + 25c_1^2c_3 + 12c_2c_3 + 38c_1c_4 + 24c_5, \quad (9)$$

reproducing eq. (17) of the note exactly. (Sanity: the coefficient of c_d is $(d-1)!$, giving 6 and 24.)

Proposition 1 of the note. $A_{(1,1,2,1)} = -1$, confirmed. It is indeed the negative coefficient of lowest total degree (degree 5).

Section 5 of the note. The Chern monomial $c_{\ell+2}^2 c_{\ell+1} c_{\ell}^2$ corresponds to the α -multiset $\{1, 1, 0, -1, -1\}$. All ten orderings with nonnegative partial sums, and their coefficients, are reproduced exactly:

α	β	A_β
$(0, 1, -1, 1, -1)$	$(0, 1, 0, 1)$	0
$(0, 1, 1, -1, -1)$	$(0, 1, 2, 1)$	1
$(1, -1, 0, 1, -1)$	$(1, 0, 0, 1)$	0
$(1, -1, 1, -1, 0)$	$(1, 0, 1, 0)$	0
$(1, -1, 1, 0, -1)$	$(1, 0, 1, 1)$	0
$(1, 0, -1, 1, -1)$	$(1, 1, 0, 1)$	0
$(1, 0, 1, -1, -1)$	$(1, 1, 2, 1)$	-1
$(1, 1, -1, -1, 0)$	$(1, 2, 1, 0)$	2
$(1, 1, -1, 0, -1)$	$(1, 2, 1, 1)$	2
$(1, 1, 0, -1, -1)$	$(1, 2, 2, 1)$	6
sum		10

Consistency: at $\ell = 0$ this monomial is $c_1 c_2^2$, whose coefficient in (9) is 10. *Section 5 of the note is correct line by line.*

4 Proved: the ℓ -free reduction

Theorem 5 (ℓ -free reduction). *For a multiset $M = \{\alpha_1, \dots, \alpha_d\}$ of integers with $\sum_i \alpha_i = 0$, define*

$$C(M) := \sum_{\substack{\text{orderings } \alpha \text{ of } M \\ \alpha_1 + \dots + \alpha_j \geq 0, 1 \leq j \leq d-1}} A_{\beta(\alpha)}, \quad \beta_j(\alpha) = \alpha_1 + \dots + \alpha_j. \quad (10)$$

Then for every $\ell \geq 0$ and every Chern monomial $\prod_{i=1}^d c_{p_i}$ with $p_i \geq 0$, the coefficient of $\prod_i c_{p_i}$ in $\text{Tp}_{A_d}^\ell$ equals $C(M)$, where $M = \{p_i - (\ell + 1)\}$. In particular that coefficient does not depend on ℓ except through M .

Proof. By (6)–(8), $\text{Tp}_{A_d}^\ell = \sum_{\beta \in \mathbb{Z}_{\geq 0}^{d-1}} A_\beta \prod_i c_{\ell+1+\alpha_i(\beta)}$. Fix a multiset $\{p_i\}$ of nonnegative integers. A term contributes to $\prod_i c_{p_i}$ iff $\{\ell + 1 + \alpha_i(\beta)\} = \{p_i\}$ as multisets, i.e. iff $\{\alpha_i(\beta)\} = M$. By Lemma 2 the β 's with a prescribed α -multiset M are exactly the orderings of M with nonnegative partial sums. Summing their A_β gives (10). The parameter ℓ appears nowhere else. $\square$

Corollary 6. *Fix d . Then:*

1. *Conjecture 3 for this d and all relative dimensions ℓ is equivalent to the single ℓ -free statement*

$$C(M) \geq 0 \quad \text{for every zero-sum multiset } M \text{ of } d \text{ integers.}$$

2. *Conjecture 3 for this d and a fixed ℓ is the restriction of the above to $\{M : \min M \geq -(\ell+1)\}$, since $p_i = \ell + 1 + \alpha_i \geq 0$.*

3. *Consequently the statements are nested: verification at ℓ implies verification at every $\ell' < \ell$.*

Remark 7. This sharpens item (4) of Section 6 of the note. “All relative dimensions ℓ ” is not an infinite family of independent conditions; increasing ℓ only admits multisets with more negative entries. It also explains why the value 10 in Section 5 is not special to $\ell = 0$: the α -multiset $\{1, 1, 0, -1, -1\}$ is the same for every ℓ , so $c_{\ell+2}^2 c_{\ell+1}^2$ has coefficient 10 for all $\ell \geq 0$. This was confirmed numerically for $\ell = 0, \dots, 6$.

Numerical confirmation of Theorem 5. Across $\ell = 0, \dots, 5$ the map $M \mapsto C(M)$ was tabulated (674 distinct α -multisets) with no inconsistency.

5 Proved: exhaustive verification for $\ell \leq 11$

Theorem 8 (verified range, $d = 5$). $C(M) \geq 0$ for every zero-sum multiset M of five integers with $\min M \geq -12$. Equivalently, Conjecture 3 holds for A_5 in all relative dimensions $\ell = 0, 1, \dots, 11$.

This is a finite exact computation, not a sample. For fixed ℓ the relevant β satisfy $\beta_j \leq (d - j)(\ell + 1)$ (from $\alpha_i \geq -(\ell + 1)$ and (7)), so truncating the chamber expansion at those per-variable caps captures every contributing monomial. Results:

ℓ	# Chern monomials	min coefficient	# negative
0	7	1	0
1	30	1	0
2	84	1	0
3	192	1	0
4	377	1	0
5	674	1	0
6	1115	1	0
7	1747	1	0
8	2611	1	0
9	3765	1	0
10	5260	1	0
11	7166	1	0

The minimum is always 1, attained at $M = (0, 0, 0, 0, 0)$ (the monomial $c_{\ell+1}^5$) and at $M = (1, 1, 1, 1, -4)$.

For $d = 4$ the chamber expansion has *no* negative Laurent coefficients in any window tested, consistent with Theorem 1 of the note; Chern positivity for $d = 4$ is therefore immediate.

How lopsided the cancellation is. At $\ell = 3$, the nine Chern monomials receiving negative Laurent input are:

Chern monomial (parts)	negative input	positive input	net
(5, 5, 4, 3, 3)	-1	11	10
(6, 6, 4, 2, 2)	-1	69	68
(6, 5, 4, 3, 2)	-1	96	95
(6, 6, 4, 3, 1)	-1	100	99
(7, 5, 4, 3, 1)	-1	345	344
(7, 6, 4, 3)	-2	668	666
(7, 6, 4, 2, 1)	-1	690	689
(8, 5, 4, 3)	-1	1402	1401
(8, 6, 4, 2)	-1	2457	2456

The count of affected monomials grows linearly (1, 3, 6, 9, 12, 15, 18 for $\ell = 0, \dots, 6$), and the tightest case is always the Section 5 family, with net 10. Within per-variable degree ≤ 9 there are

only 20 negative Laurent coefficients, with values in $\{-1, \dots, -6\}$, against positive coefficients as large as 462722. **The cancellation in Section 5 is not delicate.**

6 Closed form for F_5 and a maximal positive factorization

Using (5), the ten factors $z_m - z_l$ normalise to $1 - z_m/z_l$ and every denominator factor $z_m + z_r - z_l$ to $1 - z_m/z_l - z_r/z_l$. The signs cancel globally, giving the following identity (verified against the raw expansion of (4) to per-variable degree 8):

$$F_5(a, b, c, e) = \frac{\left[\prod_{m < l} \left(1 - \frac{z_m}{z_l} \right) \right] (1 - 2abce - bce) P}{D}, \quad (11)$$

where

$$\prod_{m < l} \left(1 - \frac{z_m}{z_l} \right) = (1 - a)(1 - ab)(1 - abc)(1 - abce)(1 - b)(1 - bc)(1 - bce)(1 - c)(1 - ce)(1 - e),$$

$$P = 1 - 2abc - bc - ce - bce + 2a^2b^2c^2e + 3ab^2c^2e + 2bc^2e,$$

$$D = (1 - 2a)(1 - 2ab)(1 - b - ab)(1 - 2abc)(1 - bc - abc)(1 - 2bc)(1 - c - abc) \\ \times (1 - 2abce)(1 - bce - abce)(1 - 2bce)(1 - ce - abce)(1 - ce - bce)(1 - e - abce).$$

Lemma 9 (Lemma 1 of the note, in the useful form). *If f, g are power series with nonnegative coefficients and zero constant term, then $\frac{1-f}{1-f-g} = 1 + \frac{g}{1-f-g}$ has nonnegative coefficients. Equivalently $\frac{1-f}{1-h}$ is coefficientwise nonnegative whenever $h - f \geq 0$ coefficientwise.*

Applying Lemma 9 greedily pairs all ten numerator factors against ten of the thirteen denominator factors:

$$F_5 = \underbrace{\frac{1-a}{1-2a} \cdot \frac{1-ab}{1-2ab} \cdot \frac{1-abc}{1-2abc} \cdot \frac{1-abce}{1-2abce} \cdot \frac{1-b}{1-b-ab} \cdot \frac{1-c}{1-c-abc} \cdot \frac{1-e}{1-e-abce} \cdot \frac{1-bc}{1-2bc} \cdot \frac{1-ce}{1-ce-bce} \cdot \frac{1-b}{1-2b}}_{\text{ten coefficientwise nonnegative factors}} \cdot \frac{(1-2abce-bce)P}{D} \quad (12)$$

$$G = \frac{(1-2abce-bce)P}{(1-bc-abc)(1-ce-abce)(1-bce-abce)}. \quad (13)$$

Each of the ten bracketed factors satisfies the hypothesis of Lemma 9 (e.g. $h-f = ab, abc, abce, bce, a, \dots$).

7 Three strategies that are ruled out

7.1 A $d = 4$ -style factorization proof cannot work

The residual G of (13) is badly negative: truncated at per-variable degree 10 it has 1382 terms of which **724 are negative**, with values ranging from -126 to 224 . So the ten stripped-off positive factors of (12) are load-bearing in the cancellation, and no regrouping of this type can make positivity manifest. (This is as it must be: Conjecture 4 is false at $d = 5$, so *no* manifestly positive factorization of F_5 exists.)

7.2 “Enumerate the negatives and dominate them” does not terminate

The negative Laurent coefficients are not a finite list, and new families appear as β_1 grows. Complete list for per-variable degree ≤ 12 (35 coefficients):

$$\begin{array}{ll}
A_{(1,1,k,k-1)} = -1, & k \geq 2, \\
A_{(2,2,k,k-2)} = -1, & k \geq 4, \\
A_{(2,2,k,k-1)} = -(k-4), & k \geq 5, \\
A_{(2,3,k,k-2)} = -1, -2, -3 & k = 10, 11, 12, \\
A_{(3,3,k,k-1)} = -4, -10, -17 & k = 10, 11, 12.
\end{array}$$

Magnitudes are unbounded and the family structure is open-ended, so pairing each negative coefficient with a dominating positive partner is an infinite program with no visible uniformity.

7.3 No closed form for $C(M)$ on simple families

Tabulating (10) along one-parameter families gives

$$\begin{array}{ll}
C(k, 0, 0, 0, -k) = 1, 10, 30, 100, 366, 1444, 6030 & (k = 0, \dots, 6), \\
C(k, 1, 0, -1, -k) = 10, 95, 344, 1401, 6126, 27955 & (k = 1, \dots, 6), \\
C(k, -1, -1, -1, 3-k) = 12, 12, 38, 24, 336, 1296 & (k = 1, \dots, 6), \\
C(k, k, 0, -k, -k) = 10, 68, 499 & (k = 1, 2, 3).
\end{array}$$

None of these factor or match a recognisable product formula. The smallest values of $C(M)$ observed (at $\ell = 6$) are $C(0, 0, 0, 0, 0) = C(1, 1, 1, 1, -4) = 1$, then $C(1, 1, 1, -1, -2) = C(1, 1, 1, 0, -3) = 5$, then a cluster of 10's.

8 The open problem, and what a proof would need

By Corollary 6, the task is:

Open. Show $C(M) \geq 0$ for every zero-sum multiset M of five integers, where $C(M)$ is the sum (10) of A_β over the ballot-type orderings of M , and A_β are the chamber-expansion coefficients of (11).

Since Conjecture 4 is false at $d = 5$, any proof must use the symmetrization essentially. Two routes that have not been excluded:

1. **A positive combinatorial model.** Exhibit a finite set $\mathcal{S}(M)$ with $|\mathcal{S}(M)| = C(M)$. The numerics in Section 7 argue against a product formula, but not against a counting interpretation.
2. **A uniform domination bound.** Show that for each M the total negative mass is strictly dominated by a designated positive ordering. The $\ell = 3$ table in Section 5 suggests enormous slack (worst case -1 against $+11$), so a crude bound might suffice. This needs control of all negative families simultaneously, not family by family.

Other things worth doing next:

- **Obtain $\mathcal{Q}_6$** and repeat everything for $d = 6$. This is the single most informative next experiment: it decides whether the negative families stay sparse and small relative to their partners, or whether Conjecture 3 itself becomes doubtful. $\mathcal{Q}_6$ is the T_6 -multidegree of $\mathcal{O}_6 \subset \widehat{N}_6$ and can be computed by Gröbner degeneration to a monomial ideal followed by summing products of normal weights over top-dimensional components (Section 1 of the note).
- **Extend Theorem 8.** The bottleneck is the dense truncated multiplication in Python ($\ell = 11$ takes ~ 48 s; cost grows roughly like ℓ^4 in the number of monomials). Rewriting the geometric-series inversion with array shifts, or using a modular/CRT scheme with bounded coefficient growth, should reach $\ell \approx 20$.
- **Prove the negative-coefficient families.** Each family $A_{(1,1,k,k-1)}$, $A_{(2,2,k,k-1)}$, $\dots$ is a diagonal of a rational function and hence satisfies a linear recurrence in k ; these are provable in closed form. Doing the same for the designated positive partners would give domination *within* each family, which is a genuine partial result even without a global proof.

9 Caveats

1. Everything rests on $\mathcal{Q}_4$ and $\mathcal{Q}_5$ in (2)–(3) being the correct Borel-orbit multidegrees. These were *not* recomputed here. The indirect evidence is: the degree constraint forces $\deg \mathcal{Q}_4 = 1$ and $\deg \mathcal{Q}_5 = 3$, both satisfied; the $d = 4$ output matches the classical Tp_{A_4} ; and the coefficient of c_d in $\mathrm{Tp}_{A_d}^{\ell=0}$ comes out as $(d-1)!$.
2. Theorem 8 is a machine verification. Its reliability is that of `thom_positivity.py`, which is calibrated against Tp_{A_4} and against eq. (17) of the note. All arithmetic is exact (Python integers); no floating point is used anywhere.
3. Theorem 5 is a genuine proof and does not depend on item 1 — it holds for any $\mathcal{Q}_d$ whatsoever.

10 Code

In the project directory:

- `thom_positivity.py` — core. Builds (4) from $\mathcal{Q}_d$, normalises factors, expands in the chamber by geometric-series inversion with per-variable truncation, and groups Laurent monomials into Chern monomials. Entry points: `laurent(d,caps)`, `chern(d,ell)`, `show(cm)`. Running it prints the $d = 4$ calibration and (9).
- `check_sec5.py` — reproduces Proposition 1 and the Section 5 table.
- `scan.py`, `high.py`, `push.py` — the ℓ -sweep behind Theorem 8.
- `structure.py` — numerical confirmation of Theorem 5.
- `negs.py` — the negative-coefficient families.
- `closed.py` — verifies the closed form (11).
- `residual.py` — computes G of (13) and its negativity.
- `family.py` — the $C(M)$ tables of Section 7.

To add a new d , enter $\mathcal{Q}_d$ in the function `Q(d,z)` in `thom_positivity.py`; everything else is generic in d .